Improving the Performance and Scalability of OpenSwathWorkflow
Contributor Proposal
Template

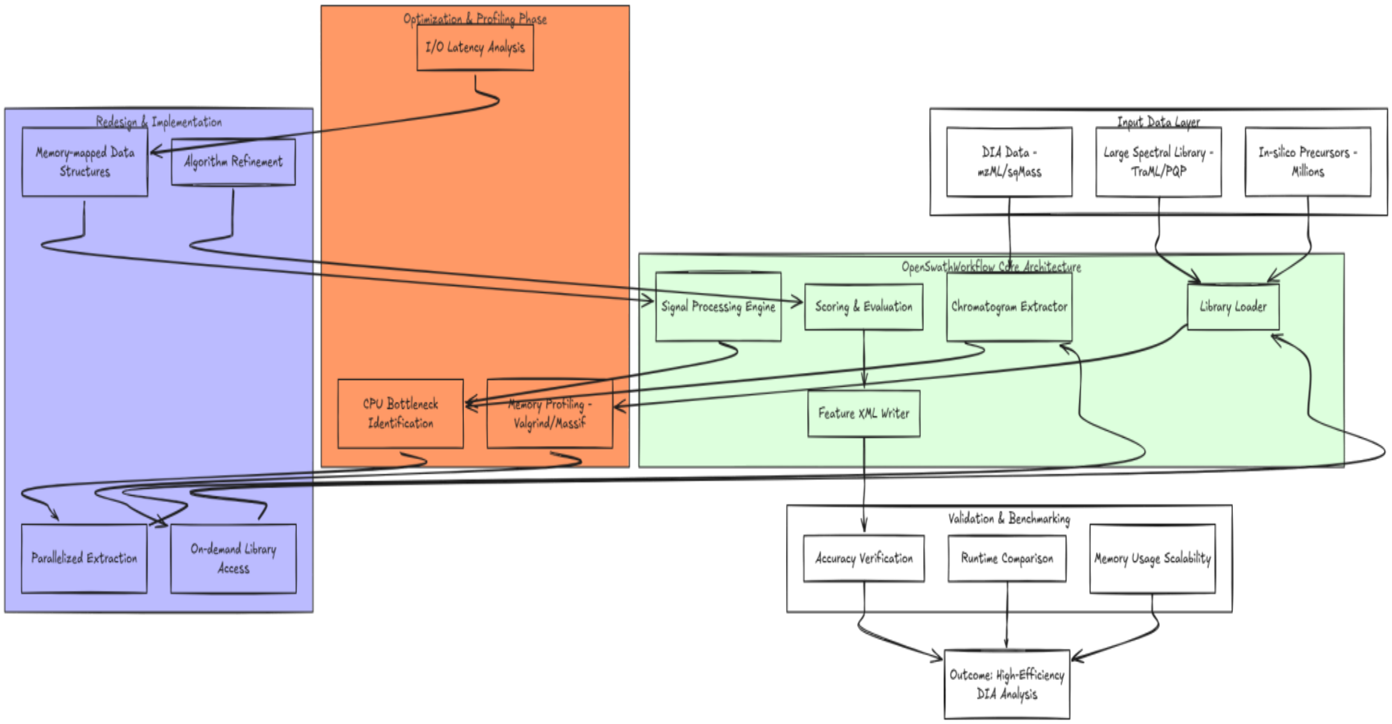
Summary : This project aims to improve the computational performance and scalability of OpenSwathWorkflow for large in silico spectral libraries. By profiling and optimizing key algorithmic components, the project will reduce runtime and memory usage while maintaining accurate peptide identification, enabling more efficient large-scale DIA data analysis.

Project Abstract

OpenSwathWorkflow is a core component of OpenMS for Data Independent Acquisition (DIA) analysis, enabling the extraction and evaluation of chromatographic signals using spectral libraries. However, the increasing use of large in silico spectral libraries containing millions of precursors introduces significant computational challenges, including high memory consumption, long runtimes, and scalability limitations.

This project aims to analyze and optimize the performance of OpenSwathWorkflow to efficiently handle large-scale spectral libraries. The work will involve identifying computational bottlenecks, improving key algorithmic components, and optimizing memory and data access patterns. The optimized workflow will be validated against the original implementation to ensure comparable accuracy while significantly improving efficiency. Ultimately, this project will enable faster and more scalable analysis of large DIA datasets.

Illustrative diagram reflecting my understanding of the **Project Abstract**, for demonstration purposes only:



Background

OpenSwathWorkflow enables accurate chromatographic signal extraction in digital proteomics (DIA). While efficient with traditional experimental libraries, modern large-scale computer libraries present computational challenges:

Memory consumption: Millions of precursors result in high memory consumption.

Increased runtime: Processing large libraries significantly increases analysis time.

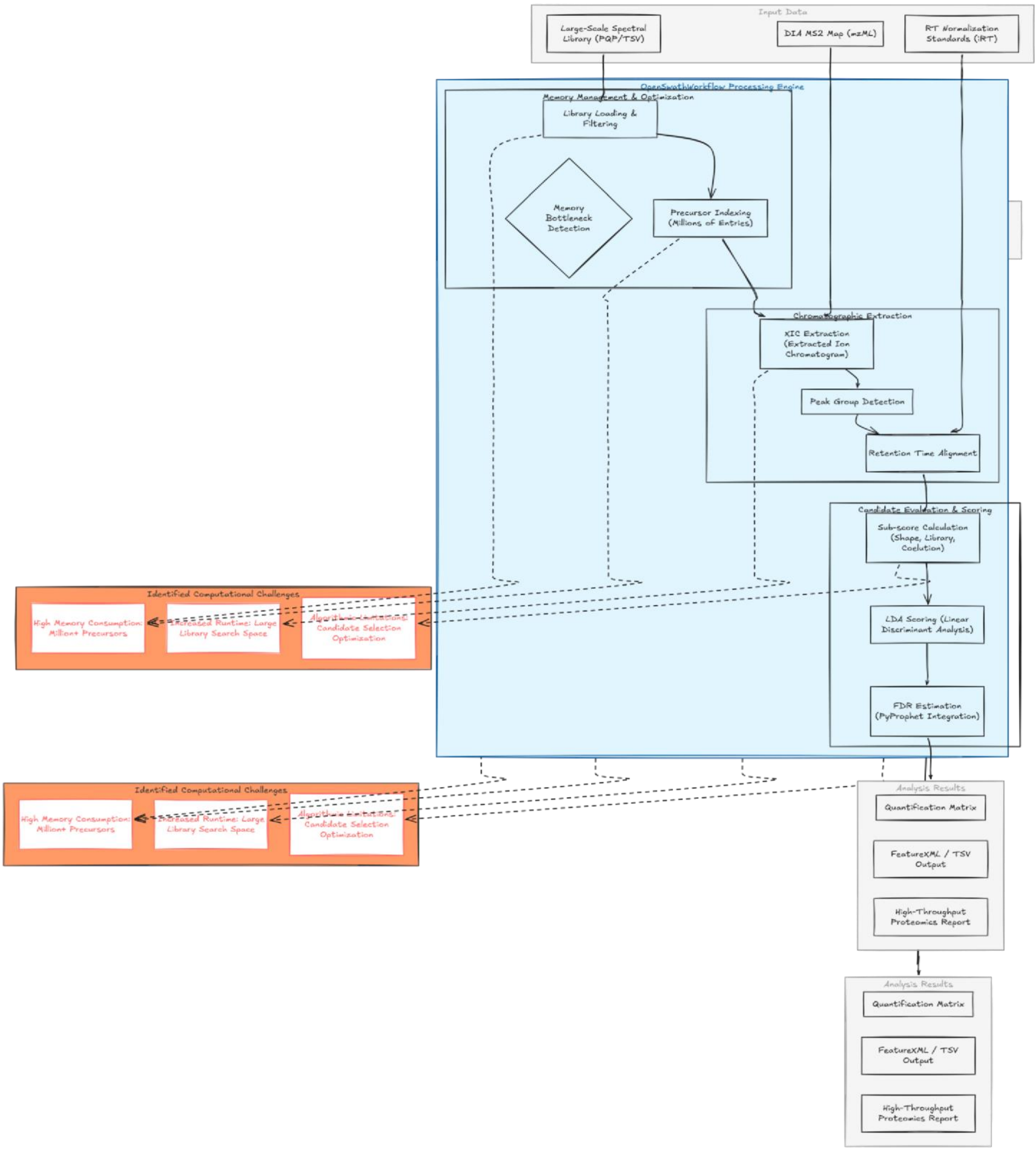
Algorithmic limitations: Candidate selection and evaluation algorithms are not fully optimized for largescale operations.

Improving OpenSwathWorkflow will benefit proteomics researchers using large computer libraries, enabling highthroughput and scalable analysis.

References:

OpenMS official documentation – (https://www.openms.de)
OpenSwathWorkflow GitHub page – (https://github.com/OpenMS/OpenMS) References for Digital Proteomics (DIA)

Illustrative diagram reflecting my understanding of the Background, for demonstration purposes only:



Design Ideas / Approach

- The project will improve OpenSwathWorkflow performance by focusing on the following: 1. **Profiling Existing Workflow:** Use benchmarking datasets to identify memory-intensive and slow components.

2. **Algorithmic Optimization:** Investigate alternative data structures, caching strategies, and scoring algorithms to reduce runtime.
3. **Parallelization:** Explore thread-level and SIMD optimizations where applicable.
4. **Validation:** Ensure that optimized workflow results remain consistent with original OpenSwathWorkflow output.
5. **Documentation & Examples:** Provide clear developer documentation for optimized components, including benchmarking results.

- **Tools & Technologies:**
- C++17/20, OpenMS codebase
- Git, GitHub
- Profiling tools (Valgrind, gprof, Perf)
- Linux/macOS/Windows build environments
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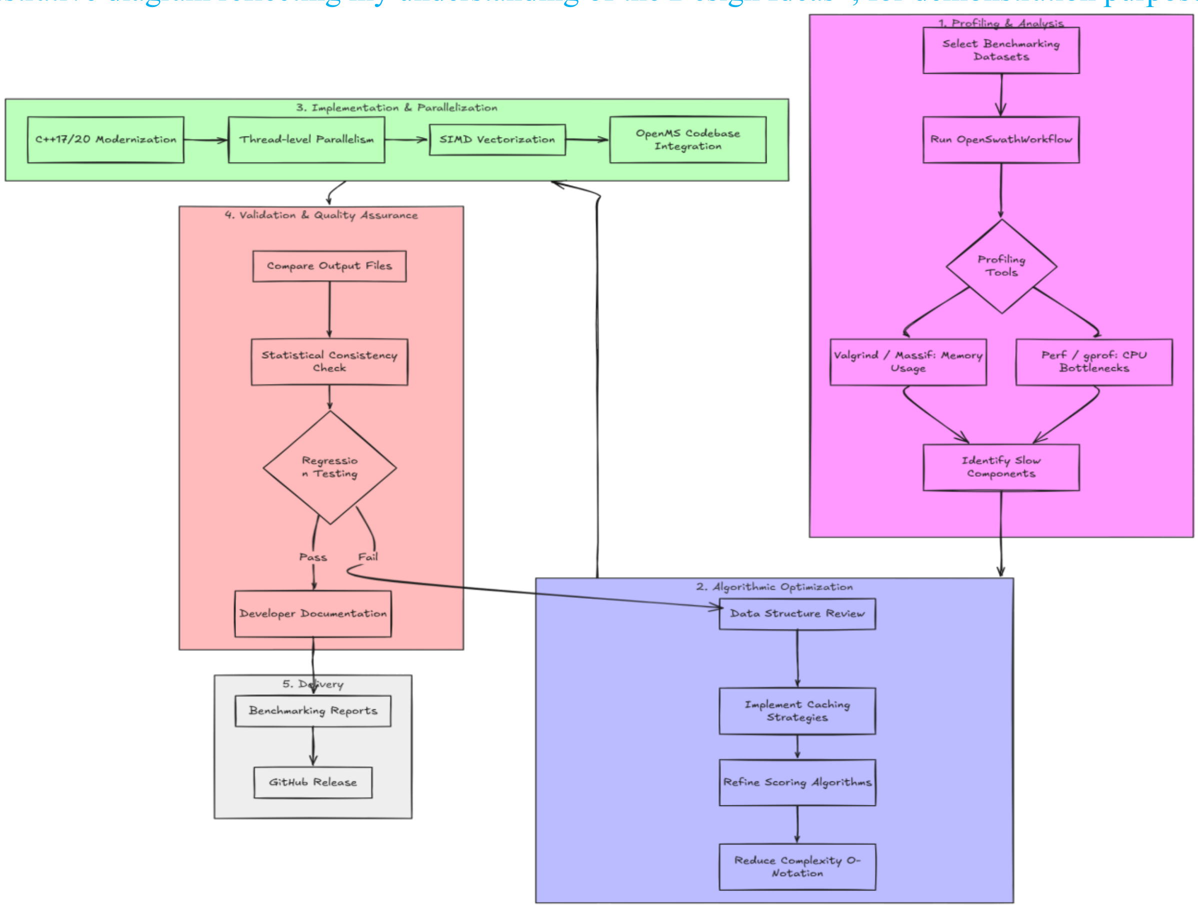
Illustrative diagram reflecting my understanding of the Design Ideas , for demonstration purposes only:



Code Affected

The exact components to be modified will be identified after an initial profiling phase of OpenSwathWorkflow.

However, the optimization effort will likely focus on key performance-critical areas, including:

- Core workflow execution logic in OpenSwathWorkflow
- Candidate selection and scoring components
- Data access and memory handling for large spectral libraries
- Computational hotspots identified through profiling

Project Goal / Impact

The primary goal of this project is to improve the performance and scalability of OpenSwathWorkflow within the OpenMS framework, enabling efficient processing of large-scale spectral libraries used in Data Independent Acquisition (DIA) analysis.

This project focuses on optimizing runtime, reducing memory consumption, and enhancing the overall efficiency of the workflow while maintaining the accuracy of peptide identification results.

For Researchers / Users

- Enable faster processing of large-scale DIA datasets.
- Reduce memory usage when working with massive spectral libraries.
- Improve productivity by decreasing analysis time.
- Maintain high accuracy while achieving better computational efficiency.

For OpenMS

- Enhance the performance of a core component used in proteomics workflows.
- Improve scalability to support modern large in silico spectral libraries.
- Strengthen OpenMS as a reliable and high-performance scientific platform.
- Provide a more efficient foundation for future algorithmic improvements.

For the Scientific & Open Source Community

- Enable high-throughput proteomics research at larger scales.
- Reduce computational cost for large biological data analysis.
- Contribute optimized, maintainable code to the open-source ecosystem.
- Support advancements in bioinformatics and data-driven biological research.

Related Work / Pre proposal Work

Before submitting this proposal, I reviewed the OpenSwathWorkflow code to understand how it handles computationally simulated large spectral libraries and identified performance bottlenecks. I also examined related issues in the OpenMS repository and considered potential optimization strategies. Previously, I developed a buffer pool manager

(https://github.com/Moemenmohamed24/Buffer_Pool_Manager)

where I efficiently managed memory, caching, and page output using the LRU-K algorithm. This experience in performance analysis, data structure optimization, and I/O optimization directly applies to optimizing OpenSwathWorkflow for large libraries. This experience gave me the confidence to propose practical improvements in algorithm efficiency and memory usage.

Schedule of Deliverables (timeline)

May 1 – May 26 (Community Bonding Period)

May 1 – May 12:

Connect with the mentor to discuss project goals. Study the architecture of OpenSwathWorkflow and understand processing of large-scale spectral libraries. Set up a local development environment with OpenMS and profiling tools.

May 13 – May 20:

Review documentation on candidate selection algorithms and scoring procedures. Explore open issues on the GitHub repository for potential optimization targets. May 21 – May 26:

Perform initial experiments on OpenSwathWorkflow to observe memory usage and runtime performance. Plan the detailed implementation for Phase I.

May 27 – July 12 (Phase I) May 27 –
June 2:

key areas requiring performance improvement using profiling. Design optimization plan for algorithms and memory management enhancements.

June 3 – June 9:
Implement initial improvements on candidate selection and scoring components. Validate results on small datasets before scaling to large in silico spectral libraries.

June 10 – June 16:
Optimize data loading and caching mechanisms. Conduct preliminary tests on large-scale in silico spectral libraries.

June 17 – June 23:
Refactor algorithms based on test results and mentor feedback. Write unit tests to ensure correctness and performance stability.

June 24 – June 30:
Document modifications and results for mentor review. Review performance improvements before finalizing Phase I.

July 1 – July 7:
Polish code, improve readability and structure. Prepare Phase I evaluation report for the mentor.
July 8 – July 12: Submit final version of Phase I for mentor evaluation.

July 13 – August 19 (Phase II)

Address any remaining mentor feedback. Further optimize performance or conduct additional tests on larger libraries.
August 20 – November 11 (Extended Timeline, if needed)
Continue additional development or optimizations. Ensure all performance improvements are stable and effective for all large datasets. Communications

Note: I have final exams in May and June, which may affect my availability for full-time work during these months. I will manage my schedule to balance study and project commitments.

Timezone/ Working hours:

Egypt Time (UTC+2), with a commitment of 15–20 hours per week and flexible working hours.

Contact Information:

- Email: moamenhamous@gmail.com
- GitHub: https://github.com/Moemenmohamed24
- LinkedIn: https://linkedin.com/in/moemen-mahmoud-337a6628b

Communication Plan:

- Weekly progress updates via email and GitHub issues
- Available for discussions on WhatsApp, Discord, Zoom, Microsoft Teams, or any platform preferred by the organization

About Me

I am Moemen Mahmoud Hamous, a computer engineering student.

I have experience in C++ and C#, system programming, algorithms, and open-source contributions.

My prior projects include database buffer managers, bank system transactions , and autonomous maze-solving bots.

I have hands-on experience with Git, GitHub, and software design patterns (OOP & SOLID principles).

My interest in performance optimization and systems programming aligns well with this project's goals of improving OpenSwathWorkflow efficiency for large-scale spectral libraries.

Prior Experience with open source

Contributed to public repositories on GitHub, mainly exploring "good first issues." Completed real-world R&D volunteer work (Grido Organization) and participated in team-based software simulations (Aspire Consulting).

Why OpenMS?

OpenMS is a widely used and highly respected open-source platform for mass spectrometry, with a strong impact on scientific research and data analysis in proteomics. OpenSwathWorkflow, as a core component of OpenMS, plays a critical role in large-scale DIA analysis, making it an exciting and meaningful area to contribute to. Improving its performance will directly benefit researchers by enabling faster and more scalable analysis of large in silico spectral libraries.
What particularly motivates me is that this opportunity represents a chance to contribute to a project that has real-world scientific impact. The work done in OpenMS goes beyond software—it supports research in biology, medicine, and data science. Being part of such an ecosystem is both inspiring and highly valuable for my growth as a software engineer.

I am especially excited that this organization is participating in Google Summer of Code, and I see this as a great opportunity to collaborate closely with experienced mentors and contribute to impactful open-source work. I am highly motivated to give my best effort, learn as much as possible, and deliver high-quality contributions that reflect positively on both the project and the GSoC program.

This project is also a perfect match for my background in C++ and system-level programming. My experience in memory management, caching, and performance optimization aligns directly with the challenges of improving OpenSwathWorkflow. Through this project, I aim not only to contribute meaningful optimizations but also to grow as a developer capable of working on largescale, performance-critical systems.

Overall, I am eager to be part of OpenMS, collaborate with its community, and contribute to making OpenSwathWorkflow more efficient and scalable while gaining valuable real-world opensource experience.